

FRUIT TEA SALES ANALYSIS AND FORECASTING

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Business Forecasting

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1. Introduction

The Fast-Moving Consumer Goods (FMCG) industry has become a global business, and the main goal of most companies is to maximize their profits. To achieve this, companies engage in marketing activities such as advertising, pricing, and other similar behaviours. It is important to use correct methods to forecast future sales so that businesses can reap the benefits (Huang et al., 2014). The following report aims to answer four questions and provide an analysis of the data, along with its findings and limitations.

2. Business background

According to Choudhury (2022), the global fruit tea market, valued at USD 2.701.3 million, is projected to grow at a CAGR of 4.3% by 2033. Fruit tea is a growing market fuelled by health-conscious consumers. Millennials and wellness enthusiasts are driving the demand. Along with diverse flavours, major industry players, such as Harney & Sons and R. Twinning, have intensified their marketing campaigns.

Currently, offline distribution channels, such as supermarkets, currently hold the majority share, but online is expected to grow rapidly.

In terms of regions, North America holds a dominant position in the market, followed by China and India. The United States comes first due to its economy, resources, and advanced technology. Furthermore, they can offer more competitive prices to their consumers, attributable to an advanced agriculture industry that provides fine quality at a low cost.

3. Data preprocessing and data exploitation

The dataset consists of 300 observations with 11 variables. After checking if there were any missing values, it was found that there were no missing values in the dataset.

Exploratory data analysis was performed using Visualisation techniques, such as bar charts and scatter plots, to gain insights into sales patterns. It included examining data distribution across different regions and visualising sales trends over months and years using the 'ggplot' library.

In table 1, correlation analyses were conducted using 'Pearson's product-moment correlation' to understand the relationships between sales and various marketing activities, including TV advertisements (Ad1..GRP.), online banner advertisements (Ad2..No_of_banners.), promotions at distribution stores (Prom..No_of_stores.), and wage percentages (Wage..Perc.).

Variables	Correlation coefficient	Marginally significant p-value	95 percent confidence interval
Ad1..GRP	0.102	0.077	[-0.011,0.213]
Ad2..No_of_banners	0.694	2.2e-16	[0.631,0.749]
Prom..No_of_stores	0.005	0.9345	[-0.109,0.118]
Wage..Perc..	0.047	0.4216	[-0.067,0.159]

Table 1 : the correlation between 'Sales units' and various factors

These findings suggest that while Ad2..No_of_banners. is strongly associated with sales units, the relationships with Prom..No_of_stores. and Wage..Perc.. appear to be negligible. After looking at correlation analysis and visualising graphs, it concluded to perform a logarithmic transformation on column 'Sales..units.' to equalise the effects of changes and increases its interpretability in relation to marketing activities. There is no transformation performed on any other columns.

This positive correlation coefficient indicates a strong positive linear relationship between the number of banners in Ad2 and the sales units. As the number of banners in Ad2 increases, there is a substantial increase in sales units.

Variations in sales can be attributed to various factors beyond marketing campaigns and efforts. Regarding the above, using the 'olsrr' package, we conducted stepwise forward regression with a significant level of 5%. The procedure introduced the variables of 'Region', 'Wage..Perc..', and 'Time'.

4. Methodology Analysis

4.1 Question 1

To model the relationship between sales and marketing activities, a linear regression model was constructed. Variables such as the number of online banner advertisements (Ad2..No_of_banners.), promotion levels (Prom..No_of_stores.), and TV advertisement effectiveness (Ad1..GRP.) were included in the model. The initial model was assessed for its fit and predictive capabilities, and residuals, the differences between the observed and predicted values, were analysed to check assumptions, including normality.

The index vs residuals graph and fitted values vs residuals graph show that points are scattered randomly, and most of them are equally distributed around the reference line. It indicates a good model fit in terms of the homoscedasticity of residuals.

A lack of any clear pattern or trend in the scatter plot of residuals vs 'Ad2' (Figure 2), residuals vs 'Ad1' (Figure 1) and residuals vs 'Promotions' (Figure 3) suggests that the linear regression model adequately captures the relationship between all three types of advertisements and the response variable (transformed_sales).

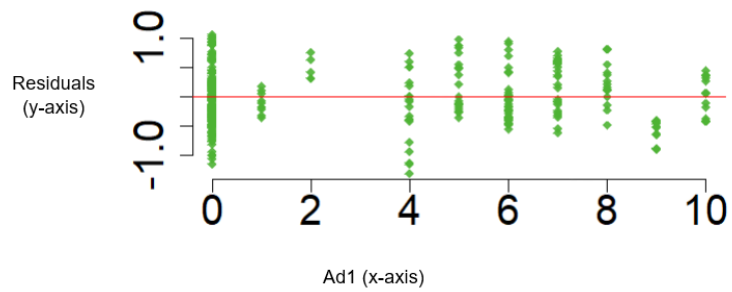


Figure 1: Residuals for Ad1

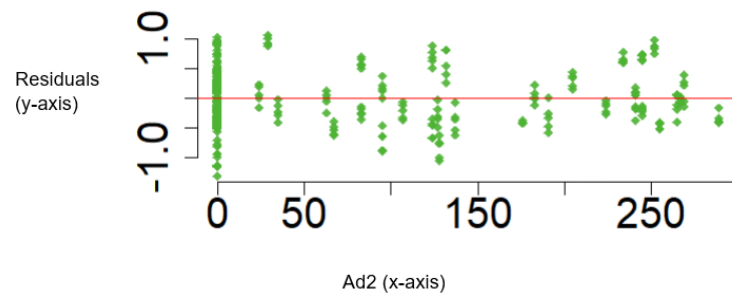


Figure 2: Residuals for Ad2

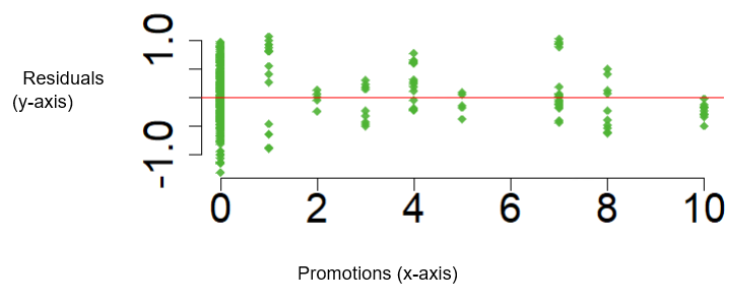


Figure 3: Residuals for Ad3

Figure 4 shows the maximum points on the straight lines, which shows that residuals adhere to the normality assumption.

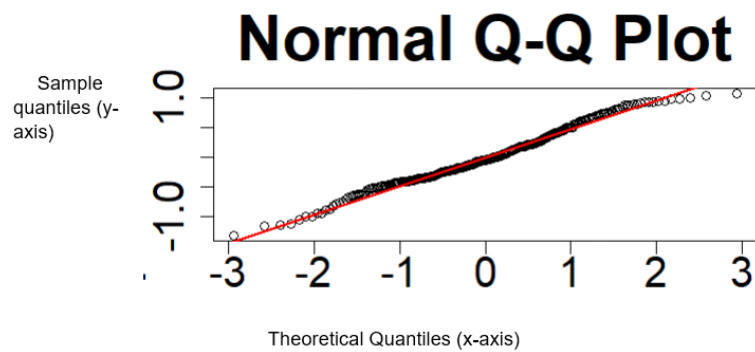


Figure 4: Q-Q plot

Figure 5 shows that the graph is slightly left skewed, indicating more negative than positive residuals. But overall, it agrees with the normal distribution assumption.

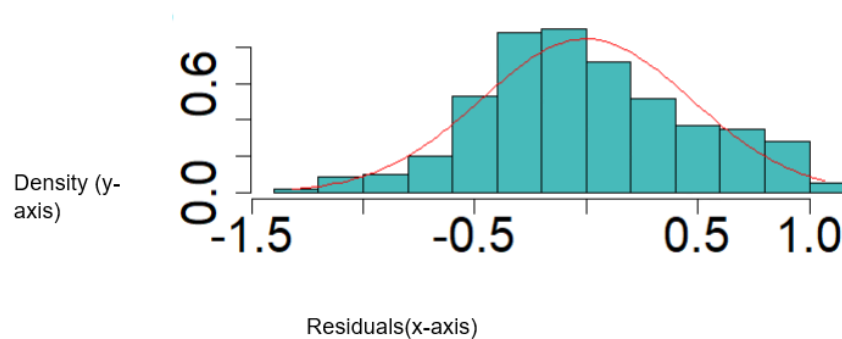


Figure 5: The histogram plot of distribution

Linear Equation:

$$\log(\text{sales}) = (1.5281787 + 0.0092097 * \text{Ad2} + 0.0063052 * \text{Promotions} + 0.0194104 * \text{Ad1} - 0.9360219 * \text{RegionSingapore} - 0.0581684 * \text{RegionSpain} - 0.8845903 * \text{RegionUK} + 0.2802360 * \text{RegionUS})$$

To account for the logarithmic transformation applied to the sales variable in the model, the coefficients are exponentiated, facilitating the interpretation of the relationship between units sold and three distinct types of advertisements.

The association between units sold and marketing activities is elucidated through exponentiated coefficients derived from a linear regression model. For Ad2..No_of_banners, a one-unit increase is associated with a 1.009252 times (or approximately 0.92%) rise in expected units sold, indicating a modest positive correlation between the number of online banner advertisements and sales. Similarly, an increase of one unit in Prom..No_of_stores is linked to a 1.006325 times (or approximately 0.63%) increase in expected units sold, reflecting a slight positive correlation with the number of promoted stores. Notably, Ad1..GRP exhibits a slightly stronger positive association, as a one-unit increase is associated with a 1.0196 times (or approximately 1.96%) rise in expected units sold, emphasising the impact of gross rating points for advertising on sales outcomes.

Advertisement	Times rise in units sold
1 unit increase in Ad1 (TV advertisements)	1.0196
1 unit increase in Ad2 (No. of banners)	1.009252
1 unit increase in Ad3 (Promotions)	1.006325

Table 2: the relationship between the increase in units sold and the application of advertising

4.2 Question 2

To determine the most efficient advertising strategy, it recommended using a linear regression function based on the results obtained from the first question. With a profit margin of 30% per sales unit, multiply the coefficient by 0.3 and divide by the cost. Comparing the two, it is evident that the result of banners is larger, indicating that using banners is the more cost-effective approach for advertising.

	Coefficient	cost	Coefficient*0.3/cost
TV ads	0.019	£2,000,000	2.911567e-09
Banners	0.009	£500,000	5.525793e-09

Table 3: the comparison of cost-effectiveness

4.3 Question 3

In general, variation in sales may not always be attributed to just marketing activities. From detailed analysis, some interesting factors can contribute to variations in the company's sales.

- Relationship between the sales and wage percentages

Notably, branches in regions with a higher wage percentile tend to achieve higher sales. Companies paying higher wages tend to have higher sales units, indicating a potential correlation between employee compensation and productivity.

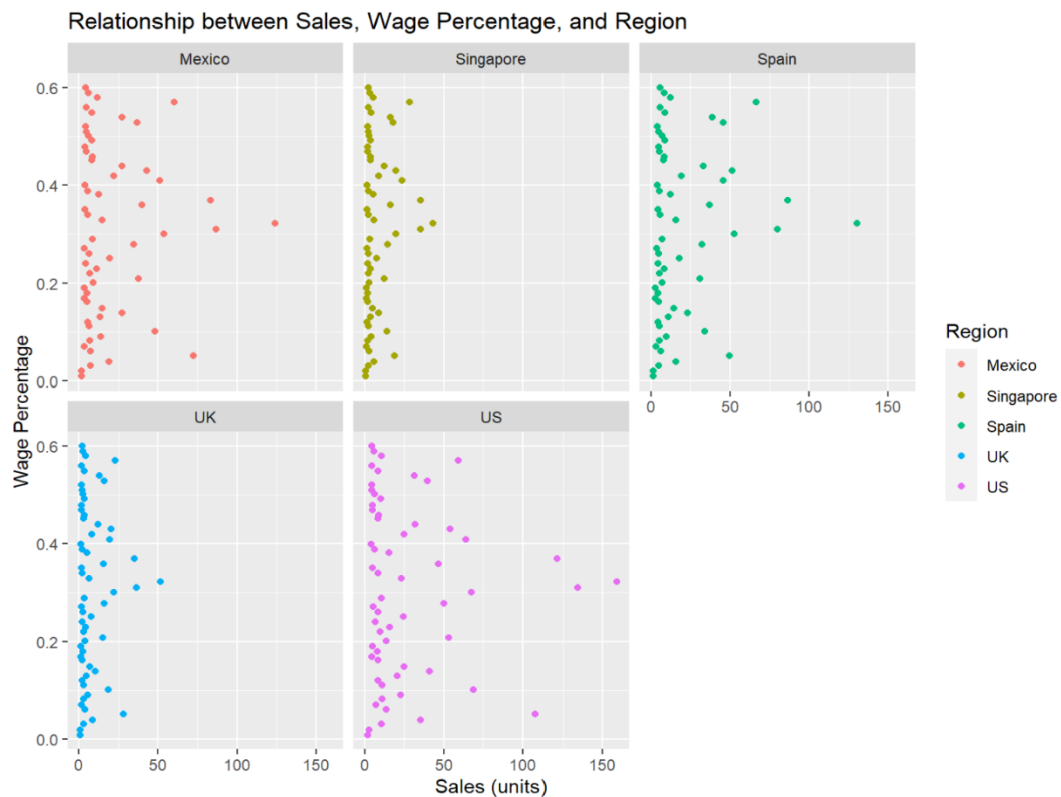


Figure 6: Relationship between sales, wage percentage, and region

- Seasonal Sales Trends

Low sales happened from November to April except January because the company added a lot of banner advertisements. Excluding the effect of advertisement, it is possible that the demand for products increases during the summer months due to weather changes.

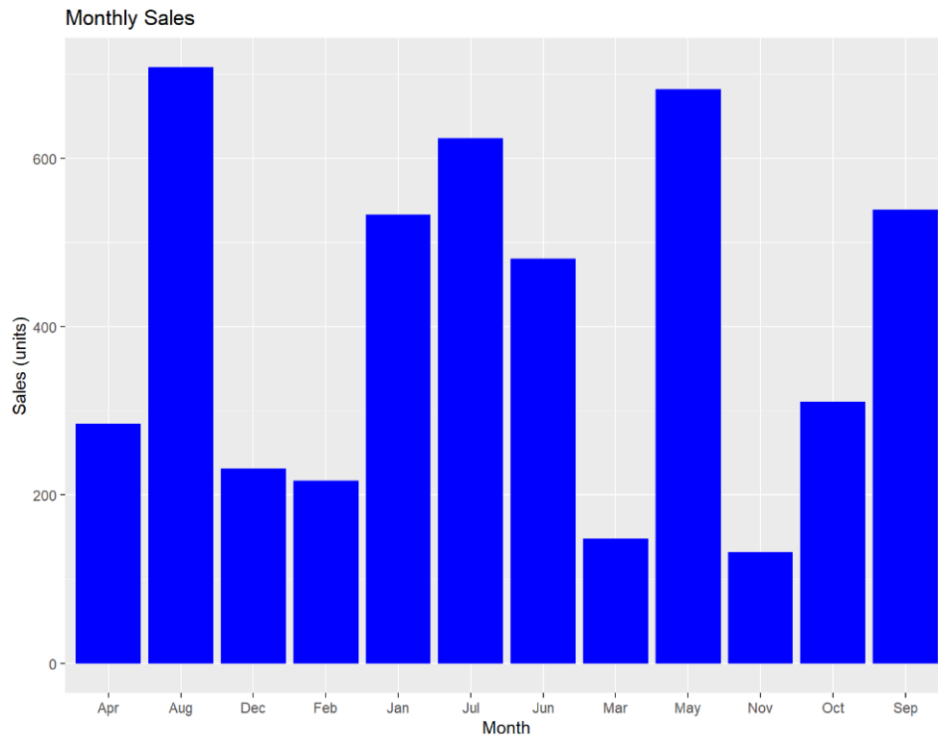


Figure 7: Monthly sales in a year

- Population Between Regions

Figure 8 depicts the sales units in different regions at various times. The US and Singapore have the highest and lowest sales, respectively, which could be attributed to their population sizes. According to Figures 9 and 10, the US had the highest population, around 320 million, between 2010 and 2014, while Singapore had the lowest population, around 5.2 million, during the same period. It is assumed that the higher the population, the greater the total consumption of necessities, leading to higher sales units. (Population Media Center, 2022).

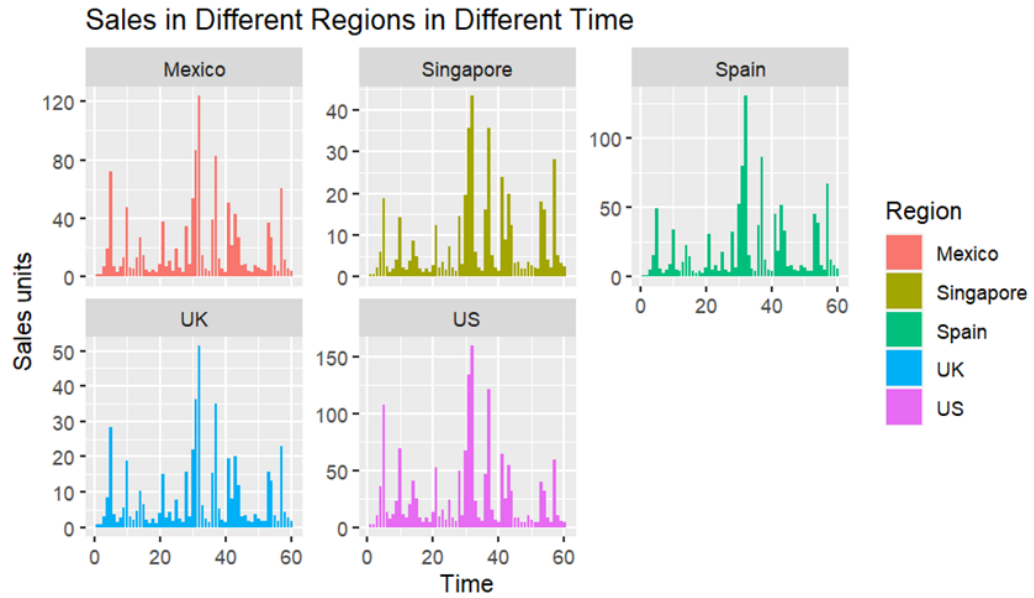


Figure 8: The sales units in each region

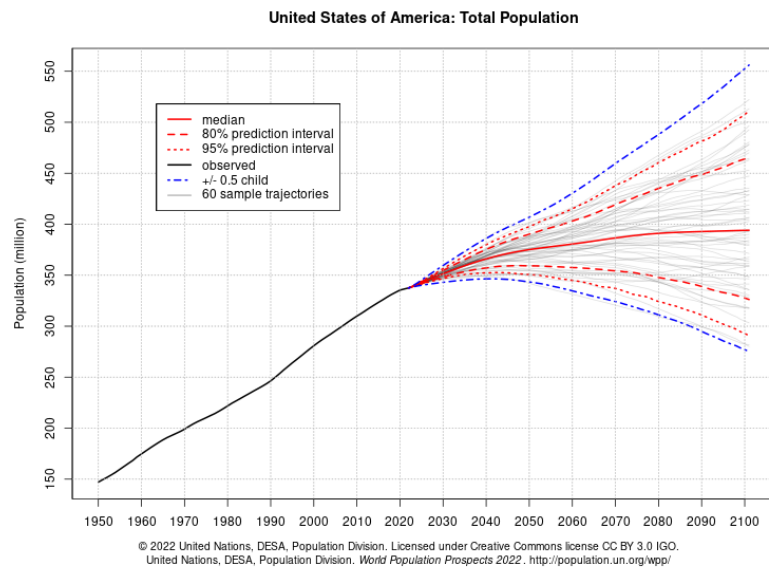


Figure 9: total population in the US

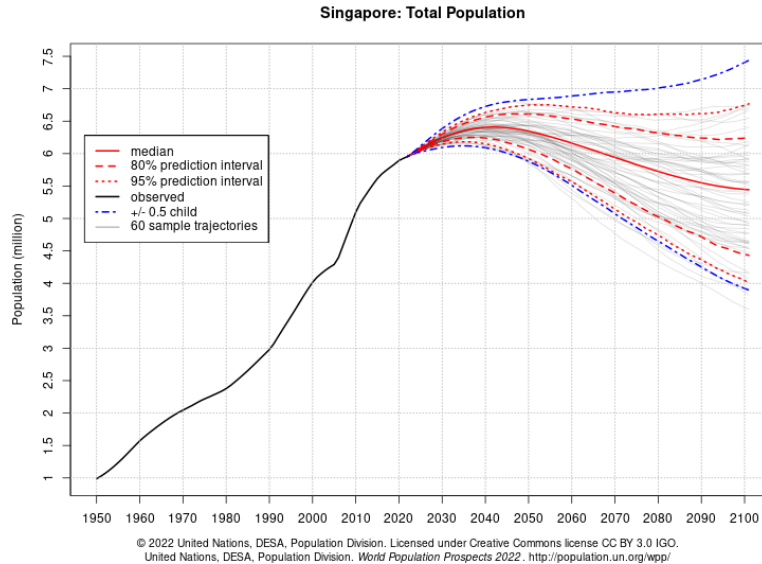


Figure 10: total population in the Singapore

4.4 Question 4

Data preprocessing required for prediction included classification of data into the 5 different regions. Further only the time-index and Sales unit information was retained, and the data was converted to a time series data.

Each time series was divided into training, validation, and test set. 4 years of data (2010-2013) was allocated to training set, 6 months (Jan/2014-June/2014) to validation set and the last 6 months (July/2014-Dec/2014) to the test set for each region time series, following an 80:10:10 ratio for train:validation:test.

Four time-series forecasting methods were tested against the validation set for namely,

1. Simple naïve forecasting
2. Naïve seasonal forecasting
3. Forecasting using decomposed seasonality and trend
4. Forecasting using decomposed remainder along with seasonality and trend

Augmented Dickey-Fuller test was performed on the remainder component for each region, Type 1 for all showed p-value < 0.5, establishing stationarity for the remainder.

The model evaluation revealed Naïve Seasonal forecasting to be the most effective model (MAPE values mentioned in table 4). Hence, Naïve seasonal forecasting model was selected and implemented on the test set.

Regions	Simple Naïve	Naïve seasonal	Seasonal+trend	Seasonal+trend+remainder
USA	610.65	64.29	94.35	101.60
Spain	92.13	48.66	197.58	204.27
UK	99.01	78.06	162.17	161.96
Mexico	748.53	146.81	291.93	292.73
Singapore	69.71	206.75	93.83	93.77

Table 4: Mean Absolute Percentage Error (MAPE) values for model validation

Figure 11 illustrates the model results for the test set in each region, and table 5 displays the final predicted values for each region.

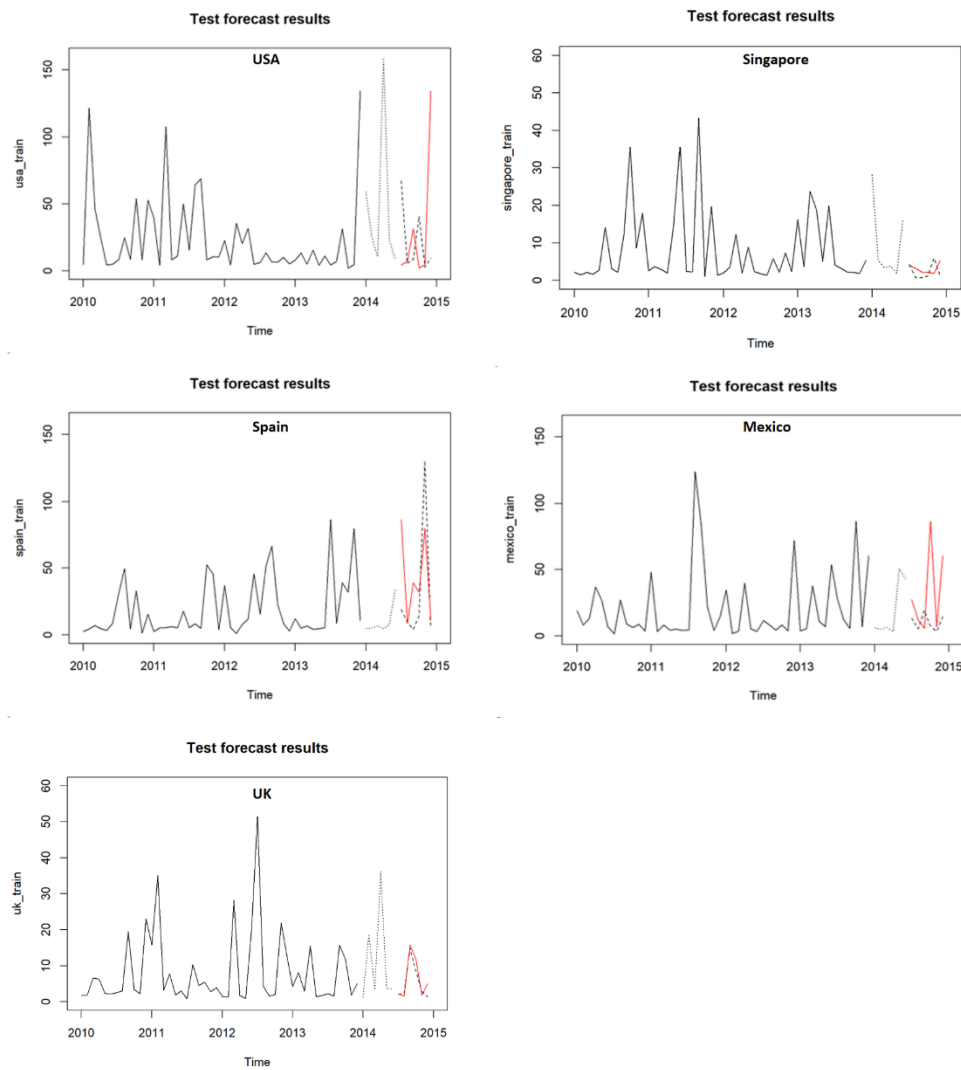


Figure 11: The forecast results for 5 regions

Regions	Predicted values
USA	610.65
Spain	92.13
UK	99.01
Mexico	748.53
Singapore	69.71

Table 5: Predicted Sales unit values for January 2015

5. Limitations

Due to a small dataset, the analysis was limited, impacting model accuracy and producing numerous zero observations. Additionally, customer behaviours were not well understood. Addressing these gaps is essential for developing better-targeted strategies.

6. Recommendations

To increase sales during winter, we need to ramp up our marketing efforts. The analysis has shown that store promotions do not have a significant impact on sales, so it would be best for the company to minimize or eliminate them to improve overall cost-effectiveness.

7. Conclusion

In conclusion, the analysis of the fruit tea market revealed significant insights. The linear regression model emphasized the positive impact of online banner advertisements and TV advertising on sales units. Cost-effectiveness comparisons underscored the efficiency of banner advertisements. Furthermore, factors such as wage percentages, seasonal sales trends, and regional population variations were identified as contributors to sales variations. The time-series forecasting model, utilizing the Naïve seasonal approach, provided reliable predictions for future sales. However, limitations arising from a small dataset and gaps in understanding customer behaviours call for further research to enhance the company's strategies.

8. Appendix 1

Table A1: USA Observed vs. Expected Sales (MAPE = 37.02%)

Time Index	Observed Units Sold	Expected Units Sold
Jan-2010	2.14	4.98
Feb-2010	3	4.61
Mar-2010	10.59	14.98
Apr-2010	35.52	26.77
May-2010	107.77	54.90
Jun-2010	13.46	4.79
Jul-2010	7.1	5.18
Aug-2010	11.13	13.25
Sep-2010	22.87	18.65
Oct-2010	68.72	43.26
Nov-2010	11.13	4.61
Dec-2010	8.25	6.36
Jan-2011	20.49	15.81
Feb-2011	40.85	24.87
Mar-2011	24.78	12.91
Apr-2011	8.25	5.21
May-2011	4.44	4.61
Jun-2011	8.08	8.99
Jul-2011	4.95	4.70
Aug-2011	13.46	14.44
Sep-2011	52.98	41.56
Oct-2011	9.58	4.61
Nov-2011	15.64	12.59
Dec-2011	6.75	5.08
Jan-2012	24.53	28.31
Feb-2012	8.58	4.73
Mar-2012	5.37	5.18
Apr-2012	49.9	57.47
May-2012	10.7	5.60
Jun-2012	67.36	53.90
Jul-2012	134.29	46.73
Aug-2012	159.17	51.73
Sep-2012	23.34	6.06
Oct-2012	8.5	4.70
Nov-2012	5	4.73
Dec-2012	46.53	46.88

Jan-2013	121.51	44.02
Feb-2013	15.49	4.82
Mar-2013	6.17	4.61
Apr-2013	4.31	4.61
May-2013	64.07	71.34
Jun-2013	24.78	11.34
Jul-2013	54.05	34.88
Aug-2013	31.82	14.44
Sep-2013	8.33	5.08
Oct-2013	8.67	8.34
Nov-2013	4.95	4.61
Dec-2013	4.85	5.18
Jan-2014	10.07	10.91
Feb-2014	6.49	5.75
Mar-2014	4.62	5.41
Apr-2014	4.35	5.18
May-2014	39.65	43.78
Jun-2014	31.5	18.27
Jul-2014	8.33	4.98
Aug-2014	4.53	4.61
Sep-2014	59.15	67.15
Oct-2014	10.49	5.18
Nov-2014	5.75	5.45
Dec-2014	4.39	5.38

Table A2: Spain Observed vs. Expected Sales (MAPE = 45.13%)

Time Index	Observed Units Sold	Expected Units Sold
Jan-2010	1.26	4.70
Feb-2010	1.38	4.35
Mar-2010	4.9	14.14
Apr-2010	15.64	25.26
May-2010	49.4	51.80
Jun-2010	6.23	4.52
Jul-2010	3.03	4.89
Aug-2010	5.21	12.50
Sep-2010	9.49	17.59
Oct-2010	34.12	40.81
Nov-2010	5.42	4.35
Dec-2010	4.53	6.00
Jan-2011	11.02	14.92

Feb-2011	23.1	23.46
Mar-2011	14.44	12.19
Apr-2011	4.71	4.91
May-2011	2.61	4.35
Jun-2011	4.53	8.48
Jul-2011	2.69	4.43
Aug-2011	7.03	13.63
Sep-2011	30.88	39.21
Oct-2011	5.21	4.35
Nov-2011	8.25	11.88
Dec-2011	4.44	4.79
Jan-2012	17.81	26.71
Feb-2012	5	4.46
Mar-2012	3.42	4.89
Apr-2012	32.14	54.23
May-2012	6.89	5.28
Jun-2012	52.46	50.86
Jul-2012	79.84	44.09
Aug-2012	130.32	48.81
Sep-2012	15.64	5.72
Oct-2012	5.64	4.43
Nov-2012	4.26	4.46
Dec-2012	36.97	44.23
Jan-2013	86.49	41.53
Feb-2013	12.06	4.55
Mar-2013	5.16	4.35
Apr-2013	4.06	4.35
May-2013	45.6	67.31
Jun-2013	19.11	10.70
Jul-2013	51.42	32.91
Aug-2013	33.12	13.63
Sep-2013	7.92	4.79
Oct-2013	8.33	7.87
Nov-2013	5.21	4.35
Dec-2013	4.76	4.89
Jan-2014	8.67	10.29
Feb-2014	6.82	5.43
Mar-2014	4.76	5.11
Apr-2014	4.14	4.89
May-2014	45.6	41.31
Jun-2014	38.86	17.24
Jul-2014	8.58	4.70

Aug-2014	5.58	4.35
Sep-2014	66.69	63.36
Oct-2014	12.18	4.89
Nov-2014	8.5	5.14
Dec-2014	5.64	5.08

Table A3: UK Observed vs. Expected Sales (MAPE = 36.11%)

Time Index	Observed Units Sold	Expected Units Sold
Jan-2010	0.94	2.06
Feb-2010	0.84	1.90
Mar-2010	2.89	6.19
Apr-2010	8.5	11.05
May-2010	28.22	22.67
Jun-2010	3.71	1.98
Jul-2010	1.49	2.14
Aug-2010	2.86	5.47
Sep-2010	5.42	7.70
Oct-2010	18.73	17.86
Nov-2010	3.03	1.90
Dec-2010	2.08	2.63
Jan-2011	4.62	6.53
Feb-2011	10.38	10.27
Mar-2011	6.62	5.33
Apr-2011	2.18	2.15
May-2011	1.28	1.90
Jun-2011	2.29	3.71
Jul-2011	1.27	1.94
Aug-2011	3.86	5.96
Sep-2011	15.03	17.16
Oct-2011	2.86	1.90
Nov-2011	4.22	5.20
Dec-2011	1.88	2.10
Jan-2012	7.77	11.69
Feb-2012	2.48	1.95
Mar-2012	1.6	2.14
Apr-2012	15.8	23.73
May-2012	3.19	2.31
Jun-2012	21.98	22.26
Jul-2012	36.23	19.29

Aug-2012	51.42	21.36
Sep-2012	6.3	2.50
Oct-2012	2.25	1.94
Nov-2012	1.6	1.95
Dec-2012	15.49	19.36
Jan-2013	35.16	18.17
Feb-2013	5.1	1.99
Mar-2013	2.08	1.90
Apr-2013	1.36	1.90
May-2013	19.49	29.45
Jun-2013	8.17	4.68
Jul-2013	20.29	14.40
Aug-2013	11.82	5.96
Sep-2013	3.03	2.10
Oct-2013	3.29	3.44
Nov-2013	1.79	1.90
Dec-2013	1.43	2.14
Jan-2014	3.53	4.50
Feb-2014	2.39	2.37
Mar-2014	1.88	2.23
Apr-2014	1.73	2.14
May-2014	15.8	18.08
Jun-2014	13.07	7.54
Jul-2014	3.42	2.06
Aug-2014	1.82	1.90
Sep-2014	23.1	27.73
Oct-2014	4.31	2.14
Nov-2014	2.56	2.25
Dec-2014	1.92	2.22

Table A4: Mexico Observed vs. Expected Sales (MAPE = 37.13%)

Time Index	Observed Units Sold	Expected Units Sold
Jan-2010	1.57	4.98
Feb-2010	1.8	4.61
Mar-2010	7.1	14.98
Apr-2010	19.11	26.77
May-2010	72.24	54.90
Jun-2010	7.32	4.79
Jul-2010	3.56	5.18

Aug-2010	7.03	13.25
Sep-2010	13.6	18.65
Oct-2010	47.94	43.26
Nov-2010	6.55	4.61
Dec-2010	5.58	6.36
Jan-2011	13.2	15.81
Feb-2011	27.11	24.87
Mar-2011	14.59	12.91
Apr-2011	5.26	5.21
May-2011	3.35	4.61
Jun-2011	5.26	8.99
Jul-2011	3.39	4.70
Aug-2011	9.12	14.44
Sep-2011	37.71	41.56
Oct-2011	6.96	4.61
Nov-2011	11.25	12.59
Dec-2011	4.14	5.08
Jan-2012	19.49	28.31
Feb-2012	6.23	4.73
Mar-2012	3.49	5.18
Apr-2012	34.47	57.47
May-2012	8.76	5.60
Jun-2012	53.52	53.90
Jul-2012	86.49	46.73
Aug-2012	123.97	51.73
Sep-2012	14.59	6.06
Oct-2012	5.58	4.70
Nov-2012	3.9	4.73
Dec-2012	39.65	46.88
Jan-2013	83.1	44.02
Feb-2013	12.3	4.82
Mar-2013	5.42	4.61
Apr-2013	3.74	4.61
May-2013	50.91	71.34
Jun-2013	21.98	11.34
Jul-2013	42.95	34.88
Aug-2013	27.11	14.44
Sep-2013	8.25	5.08
Oct-2013	8.41	8.34
Nov-2013	4.44	4.61
Dec-2013	3.71	5.18
Jan-2014	8.33	10.91

Feb-2014	6.11	5.75
Mar-2014	4.76	5.41
Apr-2014	4.01	5.18
May-2014	36.97	43.78
Jun-2014	27.39	18.27
Jul-2014	8.25	4.98
Aug-2014	4.44	4.61
Sep-2014	60.34	67.15
Oct-2014	11.59	5.18
Nov-2014	5.87	5.45
Dec-2014	4.31	5.38

Table A5: Singapore Observed vs. Expected Sales (MAPE = 43.23%)

Time Index	Observed Units Sold	Expected Units Sold
Jan-2010	0.76	1.95
Feb-2010	0.67	1.81
Mar-2010	2.16	5.88
Apr-2010	5.87	10.50
May-2010	18.73	21.53
Jun-2010	2.59	1.88
Jul-2010	1.16	2.03
Aug-2010	2.12	5.20
Sep-2010	4.26	7.31
Oct-2010	14.15	16.96
Nov-2010	2.39	1.81
Dec-2010	1.67	2.50
Jan-2011	3.78	6.20
Feb-2011	8.67	9.75
Mar-2011	5.05	5.07
Apr-2011	1.9	2.04
May-2011	1.08	1.81
Jun-2011	1.93	3.52
Jul-2011	1.16	1.84
Aug-2011	2.89	5.66
Sep-2011	12.43	16.30
Oct-2011	2.34	1.81
Nov-2011	3.56	4.94
Dec-2011	1.75	1.99
Jan-2012	7.39	11.10
Feb-2012	2.25	1.85

Mar-2012	1.43	2.03
Apr-2012	14.59	22.54
May-2012	3.16	2.20
Jun-2012	19.69	21.14
Jul-2012	35.52	18.33
Aug-2012	43.38	20.29
Sep-2012	5.99	2.38
Oct-2012	2.2	1.84
Nov-2012	1.49	1.85
Dec-2012	16.12	18.39
Jan-2013	35.52	17.26
Feb-2013	5.31	1.89
Mar-2013	2.25	1.81
Apr-2013	1.4	1.81
May-2013	23.81	27.98
Jun-2013	8.94	4.45
Jul-2013	19.89	13.68
Aug-2013	12.3	5.66
Sep-2013	3.46	1.99
Oct-2013	3.71	3.27
Nov-2013	2.08	1.81
Dec-2013	1.97	2.03
Jan-2014	3.71	4.28
Feb-2014	2.75	2.26
Mar-2014	2.18	2.12
Apr-2014	1.93	2.03
May-2014	17.99	17.17
Jun-2014	16.28	7.17
Jul-2014	4.1	1.95
Aug-2014	2.18	1.81
Sep-2014	28.22	26.34
Oct-2014	5.26	2.03
Nov-2014	3.22	2.14
Dec-2014	2.41	2.11

Overall MAPE = 39.72%

9. Appendix 2

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